

Provider	Model	Release Date	Knowledge Cutoff	Cost (Input/Output per 1M tokens)	Context Window	Type	Best Use Case
OpenAI	GPT-5.5	Apr 23, 2026	Dec 2025	\$5 / \$30 (Pro: \$30/\$180)	1M tokens	Multimodal, agentic reasoning	Long-horizon agentic coding, computer use, complex knowledge work, research assistance
Anthropic	Claude Sonnet 5	2026 (current gen)	Jan 2026	\$3 / \$15 (promo \$2/\$10 through Aug 2026)	200K (1M beta)	Multimodal, reasoning + tool use	Balanced everyday coding, writing, and agentic tasks with strong cost/performance
Google DeepMind	Gemini 3.1 Pro	~Mar 2026	Late 2025	\$2.00 / \$12.00 (≤200K tokens)	1M tokens	Multimodal reasoning	Large-document analysis, multimodal understanding (text/audio/image/video/code), enterprise agentic workflows
xAI	Grok 4.3	Apr 30, 2026	Dec 2025	\$1.25 / \$2.50	1M tokens	Multimodal, real-time data	Real-time/X-data-aware tasks, native file generation (PDF/PPTX/XLSX), general assistant use
Meta	Llama 4 (Scout / Maverick)	Apr 2025 (just outside window; still current open flagship)	Aug 2024	Open-weight — hosted from ~\$0.08–\$0.35 (Scout) / ~\$0.15–\$0.60 (Maverick)	Scout: 10M / Maverick: 1M	Open-weight, natively multimodal MoE	Self-hosted long-document retrieval (Scout) or multimodal chat/vision apps at low cost (Maverick)
Mistral AI	Mistral Large 3	Dec 2, 2025	2025	\$0.50 / \$1.50	256K tokens	Open-weight	GDPR-conscious/EU-hosted enterprise use, multilingual content, self-

						(Apache 2.0) MoE	hostable generalist workloads
DeepSeek	DeepSeek V4 (Pro / Flash)	Apr 24, 2026 (preview; official GA mid-Jul 2026)	2025	Pro: \$0.435/\$0.87; Flash: \$0.14/\$0.28	1M tokens	Open-weight (MIT) MoE	Extremely cost-efficient agentic coding and long-context reasoning, self-hosting at scale
Alibaba	Qwen 3.7 Max	May 19–20, 2026	2025/2026	\$2.50 / \$7.50 (as low as \$1.25/\$3.75 via some gateways)	1M tokens	Proprietary, agent-first reasoning	Long-horizon autonomous agents, tool-heavy workflows, drop-in replacement for Claude Code via Anthropic-compatible API
Amazon	Nova 2 Lite / Nova 2 Pro	Dec 2, 2025 (Lite GA; Pro preview)	2025	Lite: \$0.30 / \$2.50	1M tokens	Reasoning model w/ adjustable "thinking"	Everyday AWS-native workloads — customer service bots, document processing, cost-sensitive business automation
Cohere	Command A+	May 19, 2026	2025/2026	Not publicly listed (contact sales)	256K tokens	Open-source MoE ("sovereign enterprise")	Regulated/sovereign enterprise deployments, private infrastructure, RAG-heavy enterprise search

Goal to understand:

- (a) Self-attention mechanism
- (b) Vector embedding
- (c) Feed forward Neural Network

Transformers process text in Large Language Models (LLMs) by converting words into Vector Embeddings, parsing relationships between them using the Self-Attention Mechanism, and processing that contextual data deeper through a Feed-Forward Neural Network to predict the next word in a sequence

a) Self-Attention Mechanism

LLM breaks into small unit converts the text into tokens

tokens are converted into embeddings

Static word tokens embeddings Converts into Contextual Meaning embeddings allows us get a meaningful representation of text that captures relationships between words .

Query(Ask for the information) Key (Provide the information) Resulting in attention score and value (help us to compute the final representation) formula

(b) Vector embedding

Vector embeddings are numerical representations of data—such as words, images, or audio—mapped as coordinates in a multi-dimensional space

Unlike traditional keyword matching, vector search looks for the *intent* and meaning behind a query, delivering highly relevant results.

Feed forward Neural Network

Like a neural network . From input to output , no loops

Input layer - Hidden layer(Loads multiple layers based on the input) - Output layer